

L10a: Monte Carlo Pricing and Sensitivity Analysis of American Options Premium

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced American options on a CRR tree. Can simulation price them? How much does the premium change when the share price, volatility, time to expiration, or interest rate changes?

Today's concepts

- Estimate and decompose the premium of American options: Longstaff–Schwartz Monte Carlo, then intrinsic value (what exercising now would pay) and extrinsic value (the rest).
- Compute the Greeks for American options: Delta, Gamma, Theta, Vega, and Rho by finite differences on a binomial tree, as coefficients of a local Taylor expansion of the premium.
- Interpret Delta and compute the probabilities of exercise and profit: Delta as the number of shares an option is equivalent to, compared with the probability of exercise, and both probabilities from simulated share prices at expiration.

Lectures and Examples

Lecture:
CHEME-5660-L10a-Lecture-American-Derivatives-Greeks-Fall-2026.ipynb

Example 1:
CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 3:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Review on your own:
CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

Company Profile: Susquehanna

Susquehanna (SIG) is a privately held trading firm whose core business is derivatives trading. Friends who traded options on the floor of the Philadelphia Stock Exchange started it in 1987.

- Options market making: it posts prices to buy and sell, so it takes the other side of its customers' contracts. In October 2026, it was the Designated Primary Market Maker for 1,359 of 5,370 Cboe option classes, the most of any firm.
- Managing the Greeks: its book is exposed to Delta, Vega, Theta, and Rho at once, and the goal is to net these across products.
- Training: interns learn options theory, and poker teaches new traders to decide under uncertainty.

Connection to today's lecture: when a customer buys a call, the firm hedges with $100\,\Delta_c$ shares and rebalances as Gamma moves Δ_c.

Monte Carlo Pricing of American Options

We simulate share price paths under the risk-neutral pricing rule $\mathbb{Q}$, then estimate the value of holding at each date by regressing the discounted cash flows from holding on the current share price across paths. Like the lattice, Longstaff and Schwartz work backward from expiration:

Initialize: given the share price S_0, the strike price K, the payoff function $V(K, S)$ of the call or put, the time to expiration T (years), the volatility σ, the risk-free rate g_y, the number of exercise dates h after today, and the number of paths M. Compute the time between exercise dates $\Delta t = T/h$ and the one-step discount factor $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$ (expiration), where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths, those with $V(K, S_i(t)) > 0$, fit the value of holding $\hat{H}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{H}_t(S_i(t))$, **then**: exercise, so $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let's price an American put by simulation and compare it with the lattice and the European premium. We apply two rules to the same simulated paths:

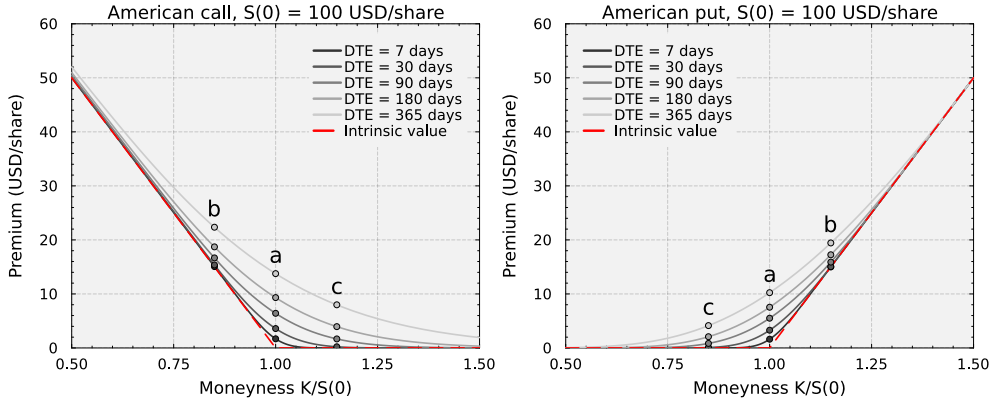
Rule at each date	Estimates
Longstaff–Schwartz: exercise when $V(K, S_i(t)) > \hat{H}_t(S_i(t))$	$\mathcal{P}^{\text{AM}}$
Hold on every path	$\mathcal{P}^{\text{EU}}$

$$\boxed{\text{early-exercise premium} \approx \mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}}}$$

Example 1: CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Extrinsic versus Intrinsic Value

The premium of an American call (left) and put (right) with $S(0)$ = 100 USD/share, priced on the CRR tree for several days to expiration (DTE), against the moneyness $K/S(0)$. Point a is at the money, b is in the money, and c is out of the money.



Intrinsic and Extrinsic Value

The premium (price) of an American option can be broken down into two components: intrinsic value and extrinsic value.

- Intrinsic value: the immediate exercise value (the red dashed line), what we would collect by exercising today, $\max(0, S(t) - K)$ for a call and $\max(0, K - S(t))$ for a put, where $S(t)$ is the current share price and K is the strike price.
- Extrinsic value: also known as time value, the rest of the premium (the gap above the red dashed line), from the potential for future price movements before expiration.

The extrinsic value $E_V(t)$ can be expressed mathematically as:

$$E_V(t) = \text{Option Price} - \text{Intrinsic Value}$$

Extrinsic Value

In expanded form, with call and put premiums $\mathcal{P}_c$ and $\mathcal{P}_p$, the extrinsic value is:

$$E_V(t) = \begin{cases} \mathcal{P}_c(K, S(t)) - \max(0, S(t) - K) & \text{(for Call Options)} \\ \mathcal{P}_p(K, S(t)) - \max(0, K - S(t)) & \text{(for Put Options)} \end{cases}$$

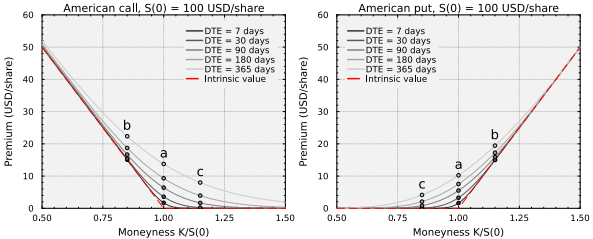
Out of the money (point c) or at the money (point a), there is no intrinsic value, so the entire option price is extrinsic value, largest at a.

As time moves towards expiration, the extrinsic value tends to decrease (time decay), because there is less time for the share price to move favorably. The points drop toward the red dashed line as the DTE shrinks.

Why do people pay money for something that is "worthless"? There is still a chance that the share price moves favorably before expiration.

Would You Ever Exercise Early?

- A call: not without dividends. Exercising buys the shares at K USD/share. Holding is better: we pay K later and earn interest on it, and we don't buy if the share price falls below K.
- A put: yes, far in the money. Exercising sells the shares at K USD/share, and collecting that cash now beats waiting.



So the call curves stay above the red dashed line (by at least the interest on K) and the put curves meet it. An American call has the same premium as a European call.

What If the Share Pays a Dividend?

A dividend changes the answer for a call. The dividend goes to those who own the shares before the ex-dividend date, not to call holders.

So if the dividend is larger than the extrinsic value we give up by exercising early, which is small far in the money, we exercise the trading day before the ex-dividend date and collect the dividend.

Review on your own: CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

The Greeks

The Greeks are local sensitivity coefficients that measure how American options premiums change with market conditions:

- Delta Δ: the change in premium for a `+1 USD/share` change in the share price.
- Theta Θ: the change in premium for a `1-day decrease` in the lifetime of the contract (time decay).
- Vega ν: the change in premium for a `+1%` change in volatility.
- Gamma Γ: the change in Δ for a `+1 USD/share` change in the share price.
- Rho ρ: the change in premium for a `+1%` change in the risk-free rate, usually minor except for longer-term options.

Most trading platforms compute these for you!

A Local Taylor Expansion of the Premium

Expanding the premium $\mathcal{P}_\star(S, \sigma, t, g_y)$ of a contract of type $\star \in \{c, p\}$ at calendar time t to second order in S and first order in the other inputs:

$$\begin{aligned}\Delta\mathcal{P}_\star &\approx \Delta_\star\,\Delta S + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2 \\ &= \underbrace{(\Delta_\star + \tfrac{1}{2}\,\Gamma_\star\,\Delta S)\Delta S}_{\text{share price}} + \underbrace{\nu_\star\,\Delta\sigma}_{\text{volatility}} + \underbrace{\Theta_\star\,\Delta t}_{\text{time}} + \underbrace{\rho_\star\,\Delta g_y}_{\text{rate}}\end{aligned}$$

$$\Delta_\star = \frac{\partial \mathcal{P}_\star}{\partial S} \qquad \nu_\star = \frac{\partial \mathcal{P}_\star}{\partial \sigma} \qquad \Theta_\star = \frac{\partial \mathcal{P}_\star}{\partial t} \qquad \rho_\star = \frac{\partial \mathcal{P}_\star}{\partial g_y} \qquad \Gamma_\star = \frac{\partial^2 \mathcal{P}_\star}{\partial S^2}$$

Derivatives are taken at the current inputs. One day is $\Delta t = 1/365$ years and one percentage point is 0.01, so over one day the premium changes by $\Theta_\star/365$, and a one-point rise in the volatility or the rate changes it by $\nu_\star/100$ or $\rho_\star/100$. Trading platforms quote these values as Theta, Vega, and Rho.

Delta and Gamma

Delta $\Delta_\star$ and Gamma $\Gamma_\star$ are the slope and the curvature of the premium as a function of the share price.

Example: a long call has $\Delta_c = 0.50$ and $\Gamma_c = 0.02$ per USD/share.

$$\Delta S = +5: \quad \Delta \mathcal{P}_c \approx (0.50 + \tfrac{1}{2}(0.02)(5)) \times 5 = 0.55 \times 5 = 2.75 \text{ USD/share}$$
$$\Delta S = -5: \quad \Delta \mathcal{P}_c \approx (0.50 - 0.05) \times (-5) = -2.25 \text{ USD/share}$$

Gamma adds $\frac{1}{2}\Gamma_c(\Delta S)^2 = 0.25$ USD/share to the Delta-only ± 2.50, so the call gains more going up and loses less going down.

Computing the Greeks of American Options

Backward induction on the CRR tree gives no formula to differentiate, so we use a central finite difference, repricing the tree with one input moved up and down by ϵ:

$$\frac{\partial \mathcal{P}_\star}{\partial x} \approx \frac{\mathcal{P}_\star(x+\epsilon) - \mathcal{P}_\star(x-\epsilon)}{2\epsilon} \qquad x \in \{S, \sigma, t, g_y\}$$

$$\Gamma_\star \approx \frac{\mathcal{P}_\star(S+\epsilon) - 2\,\mathcal{P}_\star(S) + \mathcal{P}_\star(S-\epsilon)}{\epsilon^2}$$

For technical reasons, the CRR tree does not estimate $\Gamma_\star$ or the other Greeks well, so we use the Leisen–Reimer tree, a modification of the CRR tree that gives better estimates of the Greeks.

Example 2: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

The Secret Lives of Delta

Besides measuring the premium change for a `+1 USD/share` move in the share price, Δ has two other interpretations:

- Share equivalent: for small share price moves, n contracts ($n < 0$ if sold) gain or lose as much as $100 \cdot n \cdot \Delta$ shares. Holding $-100 \cdot n \cdot \Delta$ shares makes the position delta-neutral, so Delta is also the hedge ratio.
- Probability of exercise: Delta is often used as an estimate of the probability that the option expires in-the-money. A call with $\Delta = 0.7$ suggests a 70% chance.

Probability of Exercise

Probability of exercise (POE): the probability that the contract finishes in-the-money at expiration $t = T$:

$$\text{POE} = \begin{cases} \mathbb{P}\left(S_T > K + 0.01\right) & \text{(for Call Options)} \\ \mathbb{P}\left(S_T < K - 0.01\right) & \text{(for Put Options)} \end{cases}$$

where S_T is the share price at expiration and K is the strike price.

Why the extra 0.01 USD/share? At expiration, OCC automatically exercises an equity option in the money by 0.01 USD/share or more. The POE uses the same threshold.

We'll check the conjecture that $\Delta \approx$ POE in Example 3.

Why Is Delta Close to the POE?

Proposition: for a call on a share with no dividends, on an h-step CRR tree (step Δt, up and down factors u and d, up probability q), Delta and the POE are sums over the same in-the-money terminal nodes ITM $= \{k : S_0 u^k d^{h-k} > K\}$, where k counts the up moves. With $q^* = q\,u\,e^{-g_y \Delta t}$, they are:

$$\Delta_c = \sum_{k \in \text{ITM}} \binom{h}{k} (q^*)^k (1-q^*)^{h-k} \qquad \text{Delta: up probability } q^*$$

$$\mathbb{Q}(S_T > K) = \sum_{k \in \text{ITM}} \binom{h}{k} q^k (1-q)^{h-k} \qquad \text{risk-neutral POE: up probability } q$$

Delta is larger than the risk-neutral POE: because $u > e^{g_y \Delta t}$, $q^* > q$. The in-the-money nodes have the most up moves, so the larger up probability puts more weight on them.

Derivation: advanced/delta_poe/CHEME-5660-L10a-Derivation-Delta-POE-Fall-2026.ipynb

When Is Delta Close to the POE?

- Volatility and time to expiration: the gap grows with both, so Delta is close to the risk-neutral POE for short-dated contracts and further off for long-dated or volatile ones.
- Our forecast and American puts: under $\mathbb{P}$, the up probability is the real-world probability p, so the POE can sit above or below Delta. Early exercise adds to an American put's $|\Delta_p|$, so the match is approximate.

Probability of Profit

Probability of profit (POP): the probability that the contract's profit at expiration is positive:

$$\boxed{\text{POP} = \mathbb{P}\left(\hat{P}_{\mathcal{C}} > 0\right)}$$

where $\hat{P}_{\mathcal{C}}$ is the profit of the contract at expiration. For the buyer of a call, the profit is the payoff minus the premium's future value, so it is positive when:

$$\begin{aligned}
\hat{P}_{\mathcal{C}} > 0 &\iff (S_T - K)^+ - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c > 0 && \text{profit of a long call} \\
&\iff (S_T - K)^+ > \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c && \text{add the premium's future value} \\
&\iff S_T - K > \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c && \text{a positive payoff needs } S_T > K \\
&\iff S_T > \underbrace{K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c}_{= S^\ddagger} && \text{the breakeven share price}
\end{aligned}$$

Probability of Profit and the Breakeven

The same steps apply for a put, except $S_T < S^\ddagger$ with $S^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$, and the seller profits exactly when the buyer does not:

$$\text{POP} = \begin{cases} \mathbb{P}\left(S_T > S^\ddagger\right) & \text{(bought a call or sold a put)} \\ \mathbb{P}\left(S_T < S^\ddagger\right) & \text{(sold a call or bought a put)} \end{cases}$$

The POP has the same form as the POE, with the breakeven share price in place of the strike price. For the buyer, the breakeven lies beyond the strike price, so the POP is smaller than the POE.

The POE and POP are forecasts, so we compute them under our belief $\mathbb{P}$: GBM with our estimates of the drift μ and the volatility σ.

Computing POE and POP by Simulation

Initialize: given the share price S_0, the strike price K, the premium $\mathcal{P}$ of the call or put, the time to expiration T (years), the drift μ and the volatility σ under $\mathbb{P}$, the risk-free rate g_y, and the number of samples M. Compute the breakeven share price $S^{\ddagger} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$ for a call, or $S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$ for a put, where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$.

1. For each sample $i = 1, 2, \ldots, M$ **do**:
   a. Draw a sample from the standard normal distribution: $Z_i \sim \mathcal{N}(0,1)$.
   b. Compute the share price at expiration: $S_i(T) \leftarrow S_0 \exp\left[(\mu - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i\right]$.
2. Compute the cumulative distribution function, the fraction of samples at or below each share price x: $\widehat{F}(x) \leftarrow \frac{1}{M} \cdot \#\{i : S_i(T) \leq x\}$.

Pseudocode: POE and POP

3. Compute the probability of exercise. If the contract is a call, **then** POE $\leftarrow 1 - \hat{F}(K + 0.01)$. Otherwise, POE $\leftarrow \hat{F}(K - 0.01)$.
4. Compute the probability of profit:
   - If we bought a call or sold a put, **then** the profit is positive above the breakeven: POP $\leftarrow 1 - \hat{F}(S^{\ddagger})$.
   - If we sold a call or bought a put, **then** the profit is positive below the breakeven: POP $\leftarrow \hat{F}(S^{\ddagger})$.

Output: the probability of exercise POE and the probability of profit POP.

With $\mu = g_y$ and σ set to the implied volatility, the same steps give the probabilities under $\mathbb{Q}$. These are the probabilities inside the market price, not our forecast.

Example 3: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Optional Advanced Material

One optional notebook extends this lecture:

Leisen–Reimer tree: advanced/leisen_reimer/ CHEME-5660-L10a-Advanced-LeisenReimer-Greeks-Fall-2026.ipynb

We ask why repricing the CRR tree can give a Gamma of zero, and how a tree built around the strike avoids it. We look at the CRR premium and its Delta against the share price, derive the up probability and the up and down factors of the Leisen–Reimer tree, and write pseudocode that computes the Greeks on it.

Notebook: lectures/week-10/L10a/advanced, also linked from the lecture notebook.

Summary

Today we priced American options by simulation, split the premium into intrinsic and extrinsic value, and introduced the Greeks.

Key takeaways

- Estimating and decomposing the American premium: Longstaff–Schwartz regresses the value of holding on the share price along simulated paths. Extrinsic value is largest at the money and grows with the time to expiration, and without dividends only a deep in-the-money put is worth exercising early.
- The Greeks quantify option price sensitivity: Delta, Gamma, Theta, Vega, and Rho are the coefficients of a local Taylor expansion of the premium, estimated by finite differences on a binomial tree.
- Delta serves three purposes: the premium change per USD/share move, the number of shares the option is equivalent to, and a call's POE computed with a slightly larger up probability. The POP has the POE's form, with the breakeven in place of the strike price.